Ricardo Perez Castillo

ricardoperez@berkeley.edu · LinkedIn: ricardoperezcastillo · +1 (510) 365-6415
GitHub: ricardo-pc · ricardo-pc.github.io · Berkeley, CA

Summary

Data scientist and ML engineer with 4+ years at McKinsey and Deloitte building production NLP pipelines, recommendation systems, and large-scale data infrastructure. M.A. in Statistics from UC Berkeley (May 2026). Comfortable owning problems end-to-end: from pipeline and modeling work to deployment and stakeholder communication.

Education

University of California, Berkeley — Aug 2025 – May 2026
M.A. in Statistics (completed May 2026) — Berkeley, CA
Coursework: Statistical Inference, Statistical Computing, Linear Models, Machine Learning, Bayesian Statistics.

Tecnologico de Monterrey, Campus Monterrey — Jan 2015 – Dec 2019
B.S. in Chemical Engineering — Monterrey, Mexico
Cumulative GPA: 91.38/100   Class Rank: 3/19

Experience

McKinsey & Company — Jul 2022 – Jul 2025
Senior Software Delivery Analyst, Data (Senior Data Engineer/Analyst) — Jul 2024 – Jul 2025
Software Delivery Analyst, Data (Data Engineer/Analyst) — Jul 2022 – Jul 2024

- Owned analytics and ML deployment end-to-end for Spendscape across U.S., Germany, and Latin America, partnering cross-functionally to define success metrics, ship data features, and surface insights that drove user adoption and client retention.
- Built a Snowflake data warehouse consolidating 16 ERP systems (SAP, Oracle, Coupa) using dimensional modeling (star schema with fact and dimension tables); automated ingestion (REST APIs, SFTP → AWS S3 → Python loaders) cut refresh time from 2–3 days to 24 hours, with a single extraction reaching 400M rows of transactional data.
- Quantified a $340–$460M raw-material price arbitrage opportunity for a U.S. tire manufacturer through forecasting and causal analysis on spend data, presenting findings directly to C-suite stakeholders and informing procurement strategy.
- Shipped an end-to-end production NLP text classification pipeline for spend categorization, writing model-generated labels directly to warehouse records and cutting manual classification from weeks to days.

Deloitte — Jan 2021 – Jun 2022
Consulting Analyst — Monterrey, Mexico

- Reduced supplier harmonization time by 60% (5 to 2 days) by building and deploying a spaCy NLP entity-resolution model trained on accounts payable and purchase-order data, processing millions of rows across U.S. client engagements.
- Improved ML categorization accuracy by 15 percentage points by retraining and redeploying a Google Cloud-hosted classification model, while maintaining ETL pipelines in Alteryx and Python across AP, PO, T&E, and P-card data sources.
- Analyzed $114B in combined spend across multiple industries, building Tableau dashboards that became the primary decision-making interface for finance and procurement stakeholders.

Projects

VouchNet: Reputation Service for AI Agents NANDAHack (MIT Media Lab) · Jul 2026
Python, FastAPI, PostgreSQL (Supabase), REST API github.com/ricardo-pc/vouchnet

- Designed and shipped a public reputation API letting AI agents rate and vet other agents before interacting, driven entirely by a machine-readable SKILL.md so a blind AI agent can operate the service using documentation alone. Diagnosed and fixed a live database permissions bug causing silent data loss, then validated the service end-to-end by testing and reviewing 10 other hackathon teams' live agent APIs.

Otomedi: Multi-Agent Healthcare Automation UC Berkeley AI Hackathon · Jun 2026
Python, FastAPI, Claude API, Deepgram, Next.js, Supabase github.com/ricardo-pc/Berkeley-AIHackathon

- Built a working end-to-end multi-agent pipeline that turns clinic voicemails into one-click approvals: a Claude-powered intake agent parses transcripts into structured requests, eligibility agents verify against patient records, and a human worker approves before any action executes. Designed with explicit safety constraints: emergencies bypass automation entirely and no clinical action runs without human sign-off.

Two-Stage Recommendation System: A Sparsity Analysis UC Berkeley, CS 289A · Spring 2026
Python, PyTorch, Bayesian Optimization, BPR, Matrix Factorization, NCF

- Trained and evaluated three recommendation systems (MF, NCF, two-stage BPR ranker) across five data density levels on MovieLens 1M (1M ratings, 6K users), using Bayesian optimization to independently tune each model; found that Bayesian-optimized MF outperformed NCF at every density level on NDCG@10, with the BPR reranker improving HR@10 by up to 1.2 points in sparse regimes.

Linguistic Predictors of Click-Through Rate in Headline A/B Tests UC Berkeley, STAT 230A · Spring 2026
Python, OLS, Cluster-Robust Inference, Feature Engineering, VADER Sentiment, A/B Testing

- Fit a cluster-robust OLS regression on 54,956 headline packages across 11,649 A/B tests from the Upworthy Research Archive; found that exclamation marks reduce CTR by 25% and numeric openings increase it by 6%, with results stable across HC3, cluster-robust, and impression-weighted specifications.

PECARN Traumatic Brain Injury Analysis UC Berkeley, STAT 214 · Spring 2026
Python, Bayesian Inference, Classification, scikit-learn

- Applied EDA, Bayesian inference, and classification models to 43,399 pediatric head trauma patients to validate and extend the Kuppermann clinical decision rule for CT triage.

Digital Water: Predicting Monterrey's Water Crisis Atos IT Challenge · 2020
Python, Time Series Forecasting, Machine Learning

- Built an ML forecasting system that predicted Monterrey's water supply crisis two years before 5 million people faced severe rationing, surfacing early warning signals from historical reservoir and consumption data.

Skills

Analytics & ML: A/B Testing, Experimentation Design, Statistical Inference, Causal Inference, Forecasting, Regression, Classification, Clustering, NLP, Recommendation Systems, Feature Engineering

Programming & Data: Python (pandas, NumPy, scikit-learn, statsmodels, PyTorch, spaCy), R, SQL, Snowflake, Microsoft SQL Server, AWS S3, ETL, Dimensional Modeling

BI & Infrastructure: Tableau, Power BI, Azure, GCP, Git, REST APIs, SnapLogic